ABSTRACT

This project presents the design and implementation of a PHP-based web application that demonstrates core back-end web development concepts. The system provides dynamic content fetched from a MySQL database and supports user registration, authentication, authorization, file upload, and full CRUD (Create, Read, Update, Delete) operations. Server-side validation is implemented to ensure data integrity and security. Session-based authentication ensures that only authorized users can access restricted features, while role-based authorization allows administrators to manage all records and normal users to manage only their own data. The project was developed using PHP, MySQL, HTML, CSS, and XAMPP, and serves as a practical demonstration of secure back-end web application development.

ACKNOWLEDGEMENTS

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ABBREVIATIONS

Abbreviation	Meaning
PHP	Hypertext Preprocessor
CRUD	Create, Read, Update, Delete
DB	Database
UI	User Interface
CSRF	Cross-Site Request Forgery

LIST OF FIGURES

Figure 1: Home Page

Figure 2: Registration Page

Figure 3: Login Page

Figure 4: Dashboard

Figure 5: Create Record Form

Figure 6: Edit Record Page

LIST OF TABLES

Table 1: Users Table Structure

Table 2: Records Table Structure

Table 3: Test Cases

CHAPTER 1: INTRODUCTION

1.1 Background

Web applications are widely used to manage and store information dynamically. PHP is a popular server-side scripting language used for building dynamic and interactive web applications. This project focuses on developing a secure PHP web application with database integration and user authentication.

1.2 Problem Statement

Many basic web applications lack proper authentication, validation, and access control mechanisms. This can lead to unauthorized access and data inconsistency. The problem is to design a secure system that allows authenticated users to manage data safely.

1.2 Objectives

- To develop a PHP-based dynamic web application
- To implement user registration and login functionality
- To apply session-based authentication and authorization
- To perform CRUD operations using MySQL
- To implement secure file upload and validation

1.4 Scope & Limitations

Scope:

- User authentication and role-based access
- Secure CRUD operations
- File upload functionality

Limitations:

- No email verification
- Basic UI design
- No REST API integration

CHAPTER 2: SYSTEM ANALYSIS & DESIGN

2.1 Requirement Analysis

Functional Requirements

- User registration and login
- Role-based authorization (Admin/User)
- Create, edit, delete, and view records
- File upload with validation

Non-Functional Requirements

- Security
- Reliability
- Performance
- Data integrity

2.2 System Architecture

The system follows a client-server architecture:

Frontend: HTML, CSS

Backend: PHP

Database: MySQL

(User → Browser → PHP Server → MySQL Database)

2.3 Database Design

Users Table

Field	Type
id	INT (PK)
name	VARCHAR
email	VARCHAR
password	VARCHAR
role	ENUM

Records Table

Field	Type
id	INT (PK)
title	VARCHAR
description	TEXT
file	VARCHAR
user_id	INT (FK)

CHAPTER 3: IMPLEMENTATION

3.1 Tools & Technologies

- PHP
- MySQL
- HTML & CSS
- XAMPP
- VS Code

3.2 System Modules

Public Module

- Home Page
- View Page
- Registration
- Login

Private Module

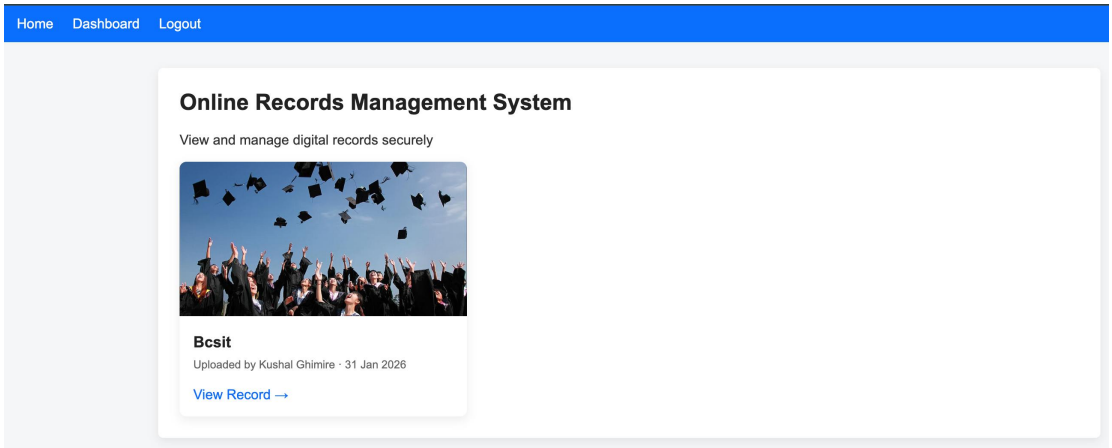
- Dashboard
- Create Record
- Edit Record

- Delete Record
- Logout

3.3 Screenshots with Explanation

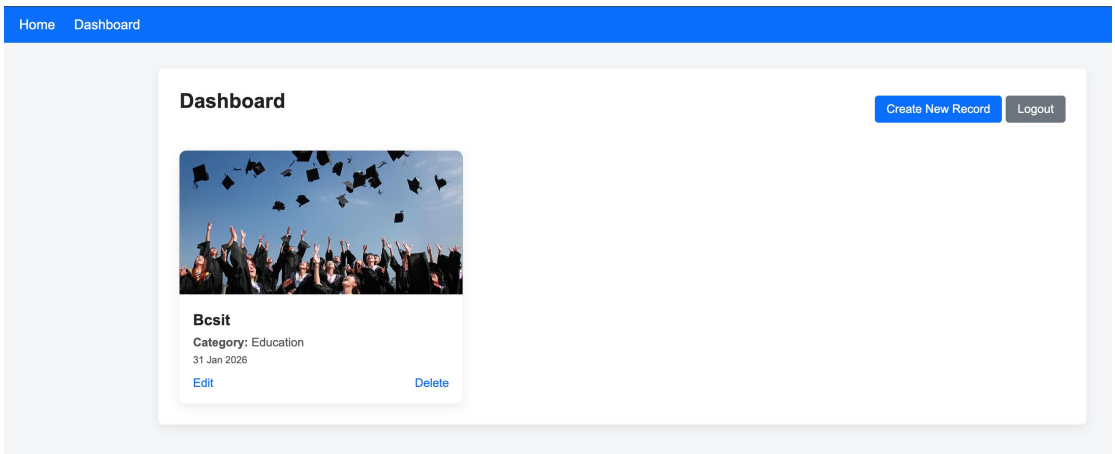
Home Page

Home page displays dynamic records from database



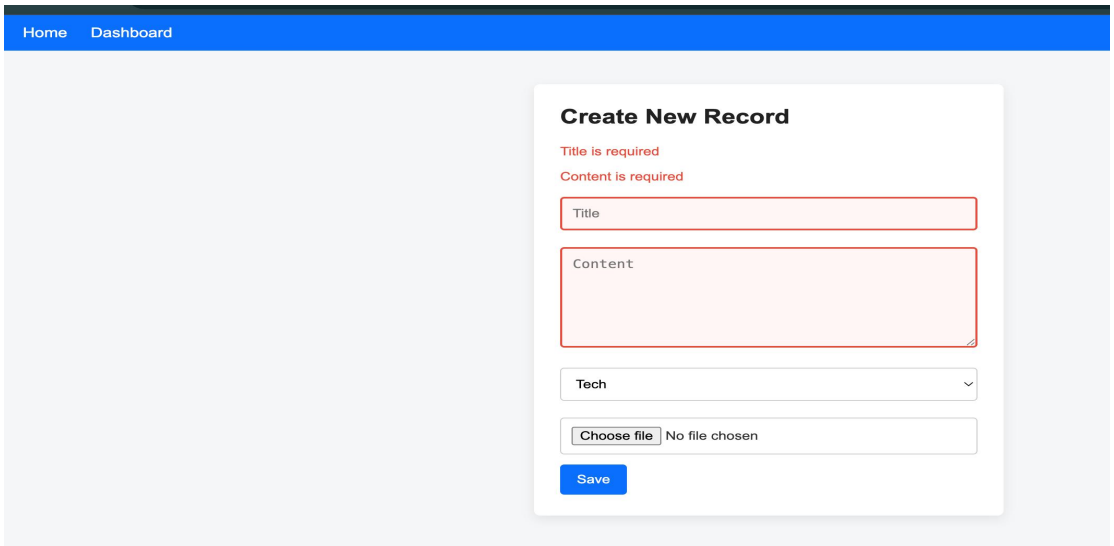
Dashboard

Dashboard shows user-specific data



Validation

Create/Edit forms validate inputs on server-side



CHAPTER 4: TESTING & VALIDATION

4.1 Test Cases

Test Case	Expected Result
Empty form submission	Error message
Invalid email	Validation error
Unauthorized access	Redirect to login
File size exceeded	Upload blocked
User edits own record	Success
User edits others record	Denied

4.2 Security Measures

- Password hashing using password_hash()
- Prepared statements (SQL injection prevention)
- Session-based authentication
- Authorization checks

- File upload validation

4.3 Error Handling

Try-catch blocks and conditional checks are used to handle runtime errors and database exceptions gracefully.

CHAPTER 5: RESULTS & DISCUSSION

The system successfully performs all CRUD operations and enforces authentication and authorization rules. Admin users can manage all records, while normal users are restricted to their own data. Server-side validation ensures reliable data entry, and file uploads are securely handled.

CHAPTER 6: CONCLUSION & FUTURE ENHANCEMENTS

6.1 Conclusion

The PHP web application meets all project objectives and demonstrates core backend development concepts. The system is secure, functional, and easy to use.

6.2 Future Enhancements

- Email verification
- Password reset feature
- Improved UI using Bootstra
- REST API integration
- Pagination and search

APPENDICES

This appendix includes important code snippets that demonstrate key functionalities of the system such as authentication, authorization, validation, and file upload security.

Appendix A: Sample Code Snippet

A.1 Password Hashing During Registration

This code ensures that user passwords are securely stored in hashed format in the database.

```
$hashedPassword = password_hash($password, PASSWORD_DEFAULT);
```

A.2 Password Verification During Login

This code verifies the user-entered password with the hashed password stored in the database.

```
if (password_verify($password, $user['password'])) {
   $_SESSION['user_id'] = $user['id'];
   $_SESSION['role'] = $user['role'];
}
```

A.3 Session-Based Authentication Check

This code restricts access to private pages for unauthenticated users.

```
if (!isset($_SESSION['user_id'])) {
   header("Location: login.php");
   exit;
}
```

A.4 Server-Side Form Validation

This snippet validates required fields on the server before inserting data into the database.

```
if (empty($title)) {
   $errors['title'] = "Title is required";
}
if (strlen($title) < 3) {
   $errors['title'] = "Title must be at least 3 characters";
}
```

A.5 File Upload Validation

This code validates file type and size before uploading.

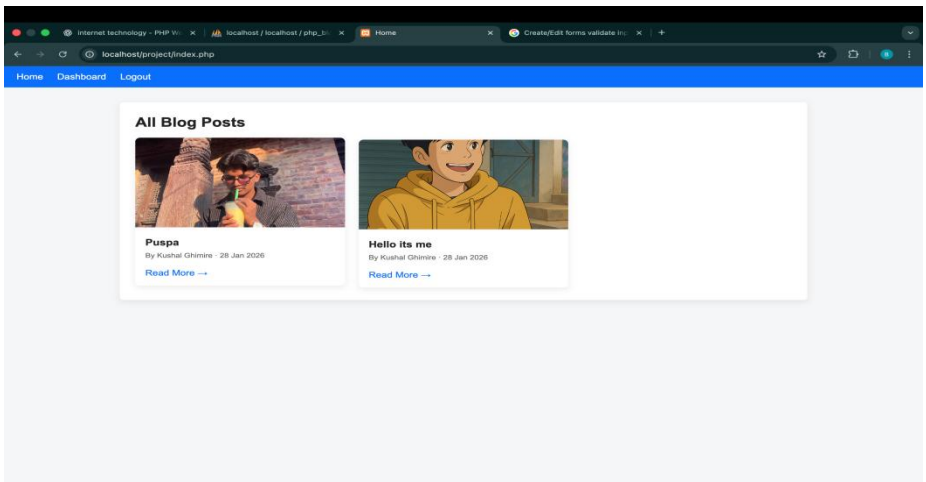
```
$allowed = ['jpg', 'jpeg', 'png', 'pdf'];
$maxSize = 2 * 1024 * 1024;

if (!in_array($fileExt, $allowed)) {
```

```
    $errors[] = "Invalid file type";
}
```

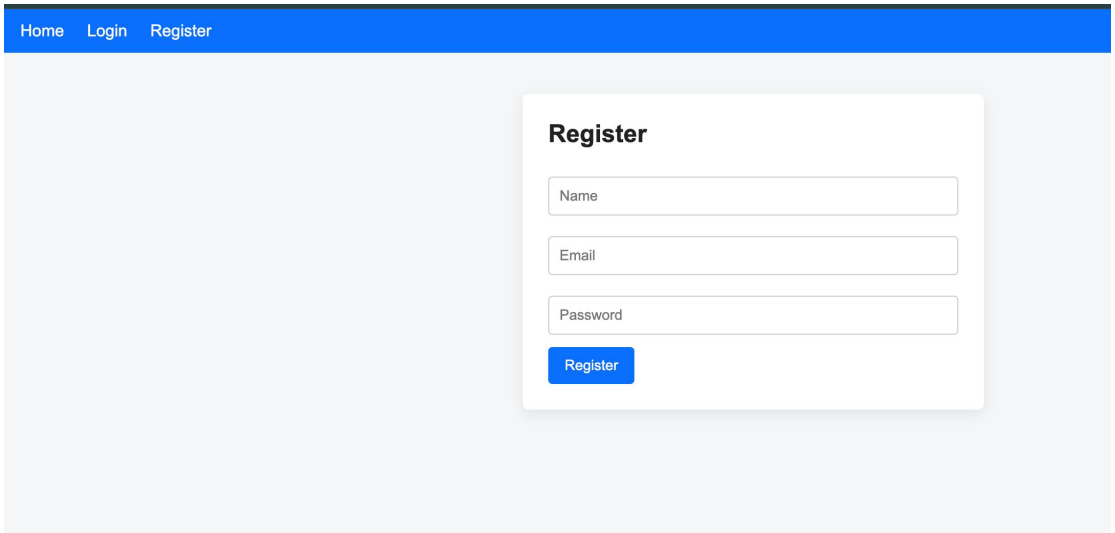
Appendix B: Screenshots

This appendix presents screenshots of the working system to demonstrate the implementation of required features.

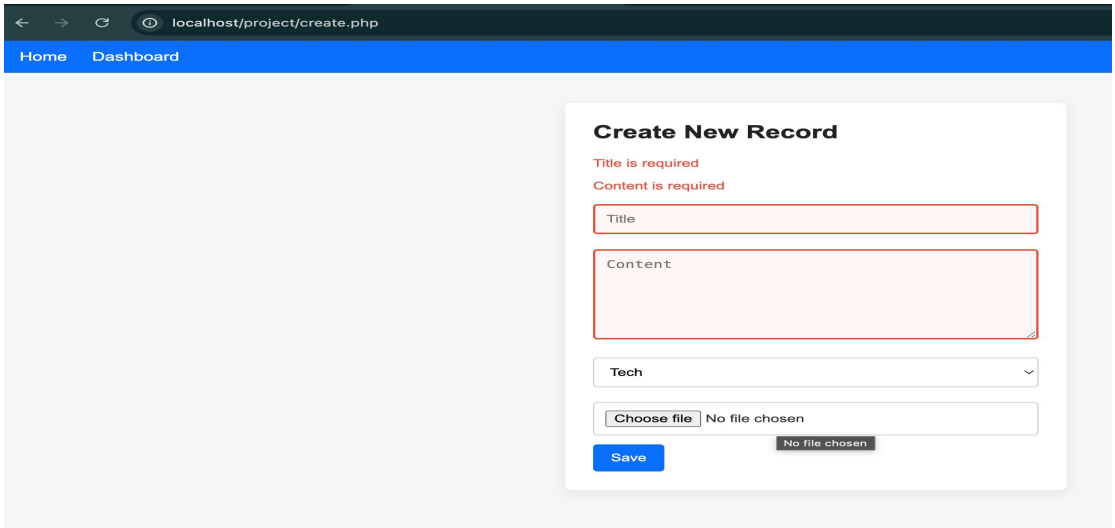
B.1 Home Page (Public View)



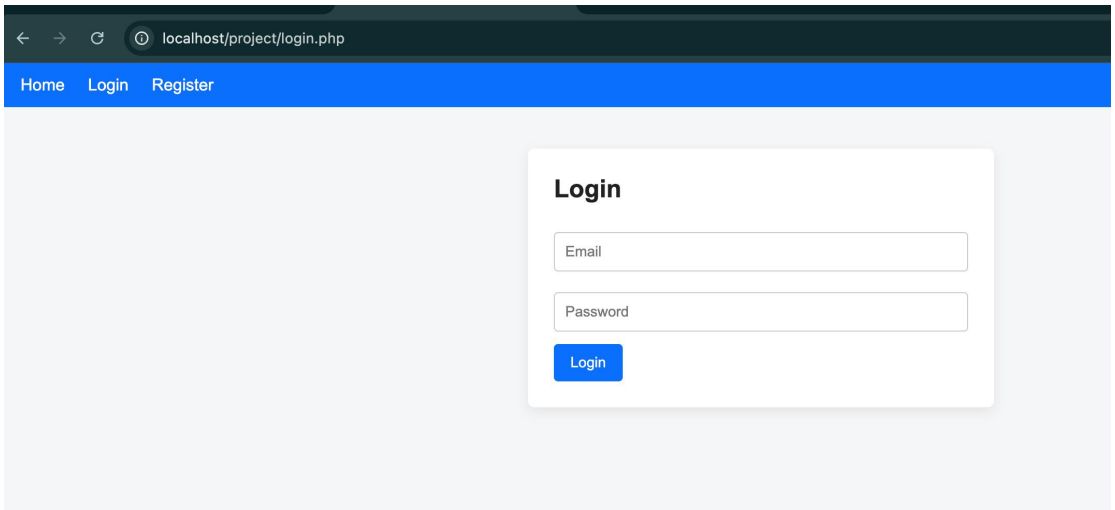
B.2 User Registration Page



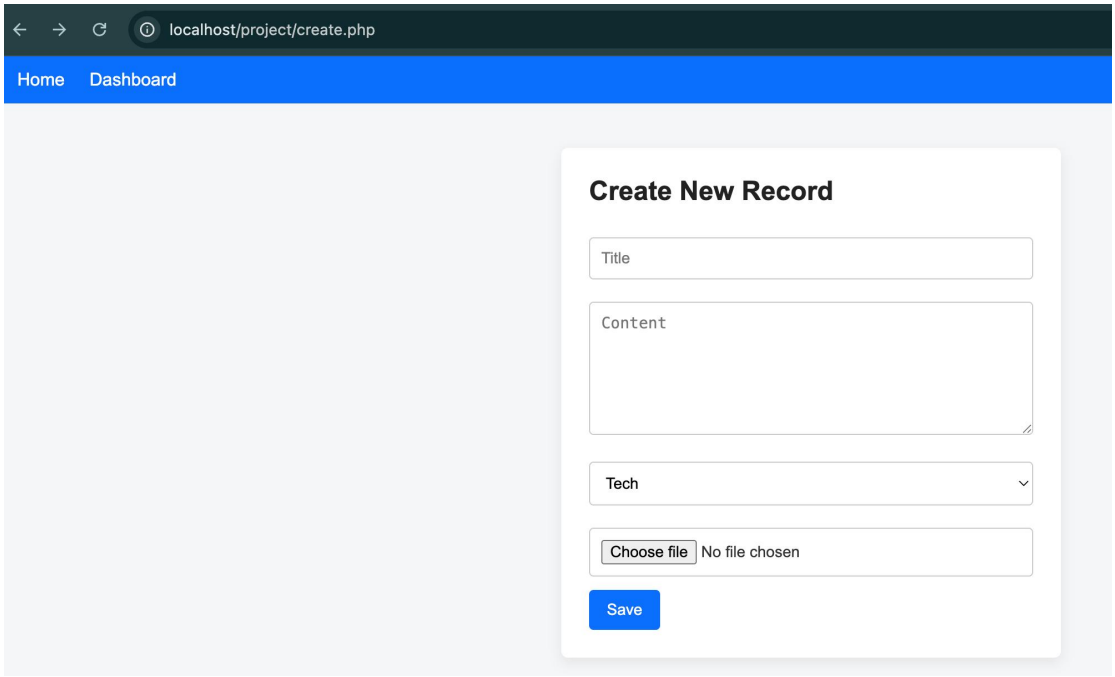
B.3 Server-Side Validation Error (Create Post Form)



B.4 Login Page



B.6 Create Post Page



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